II.D Principles of Flight

Lesson Plan. Area of Operation II, Technical Subject Areas. ACS Task code AI.II.D.

[Name], CFI Applicant

Ground lesson. Uses the matching slide deck "II.D Principles of Flight - Slides" (49 slides).

1. Objective

The student will explain how an airfoil produces lift, the four forces and types of drag, airplane stability, turning tendencies, load factors, and wingtip vortices, apply them to the C172S, and manage the risks that come from them.

ACS objective: To determine the applicant understands aerodynamics appropriate to the desired instructor certificate, can apply that knowledge, manage associated risks, demonstrate appropriate skills, and provide effective instruction.

2. Elements

ACS code	Element	Slides
AI.II.D.K1	Airfoil design characteristics.	4, 5, 6, 7, 8, 9, 10, 11, 12
AI.II.D.K2	Airplane stability, maneuverability and controllability.	24, 25, 26, 27, 28, 29, 30, 31
AI.II.D.K3	Turning tendency (e.g., torque, p-factor, spiraling slipstream, and gyroscopic precession).	32, 33, 34, 35, 36
AI.II.D.K4	Forces acting on an airplane.	13, 14, 15, 16, 17, 18
AI.II.D.K5	Load factors in airplane design.	37, 38, 39, 40, 41, 42
AI.II.D.K6	Wingtip vortices and appropriate precautions.	19, 20, 21, 22, 23
AI.II.D.R1	The basic aerodynamic principles of flight.	43, 44, 45
AI.II.D.S1	Deliver instruction on principles of flight, including at least three of the elements listed in K1 through K6.	46

3. Schedule

Part	Slides	Time
Introduction	Slides 1 to 3	5 min
Airfoil design and lift (K1)	Section 1	15 min
Four forces and drag (K4)	Section 2	10 min
Wingtip vortices (K6)	Section 3	8 min
Stability and control (K2)	Section 4	12 min
Turning tendencies (K3)	Section 5	8 min
Load factors (K5)	Section 6	10 min
Risk, teaching plan, review	Final slides	10 min
Total		78 min

4. Equipment

- Slide deck
- Whiteboard (the applicant's Principles of Flight board works as the summary)

- Model airplane and a sheet of paper (Bernoulli demo)
- C172S POH
- PHAK Chapters 4 and 5
- AIM Chapter 7, Section 4

5. Introduction

Attention: Ask: "Have you ever stuck your hand out of a car window and felt it lift?" Every principle today starts from that feeling.

Motivation: Stalls, spins, wake encounters, and overstressed airframes all come from these principles. Understanding them makes every maneuver make sense and keeps you out of the accident reports.

Overview: We go in a chain: how the wing makes lift, the forces and drag that come with it, the vortices drag leaves behind, why the airplane stays stable, what tries to turn it, and how much load it can take. Then we manage the risks and plan how to teach it.

6. Development

Teach each slide in order. Slide numbers match the deck exactly.

Slide 1: Principles of Flight

ACS: AI.II.D

Attention step: ask the student if they have ever put a hand out of a car window and felt it lift when they tilted it. The PHAK uses this same example (PHAK 4-10). Today we explain why.

Source: PHAK Fig 5-1

Slide 2: Objective

ACS: AI.II.D

State the objective: the student will understand airfoil design, the four forces, stability, turning tendencies, load factors, and wingtip vortices, and will know how to manage the risks that come from them.

Source: ACS II.D

Slide 3: Lesson Roadmap: Each Topic Builds the Next

ACS: AI.II.D

Show the roadmap. The order is chosen so each topic sets up the next: the wing makes lift; lift comes with forces and drag; induced drag makes wingtip vortices; then we ask why the airplane stays stable, what pulls it off course (turning tendencies), and how much load it can take. Risk management ties it all together.

Teaching the order this way follows the AIH idea of building from the known to the unknown and connecting each new idea to the last one.

Step	Topic	Question that leads to the next
1	Airfoil design and lift (K1)	The wing makes lift. What other forces act on the airplane?
2	Four forces and drag (K4)	Induced drag comes from lift.

		What does it leave behind?
3	Wingtip vortices (K6)	We have forces in balance. Why does the airplane stay that way?
4	Stability, maneuverability, controllability (K2)	What forces try to pull it off its path?
5	Turning tendencies (K3)	When we maneuver, how much load can the airplane take?
6	Load factors (K5)	Now manage the risks (R1) and teach it (S1).

Transition: Start with step 1: how the wing makes lift.

Source: ACS II.D

Slide 4: Airfoil Design and Lift

ACS: Step 1 of 6 | AI.II.D.K1

Section start: Airfoil Design and Lift. Goals: Parts of an airfoil; Bernoulli and Newton; Relative wind and AOA; Critical AOA and the stall.

Source: PHAK 4-6 to 4-8, PHAK 5-4

Slide 5: Definitions: Chord Line and Camber

ACS: AI.II.D.K1

Read the definitions. The rounded front is the leading edge; the narrow, tapered back is the trailing edge (PHAK 4-6). The mean camber line is equidistant from the upper and lower surfaces (PHAK 4-7).

The glossary adds that the curved upper camber makes the airflow above the wing much faster than below it (PHAK G-6).

Source: PHAK 4-6, PHAK G-6, PHAK Fig 4-5

Slide 6: Airfoil Design Characteristics

ACS: AI.II.D.K1

An airfoil is shaped to take advantage of the air's response to physical laws, giving a positive pressure lifting action from below and a negative pressure lifting action from above (PHAK 4-7).

Figure 4-6 shows the progression from early airfoils to laminar flow and supersonic shapes. Each design is chosen for the airplane's purpose.

Source: PHAK 4-6, PHAK 4-7, PHAK Fig 4-6

Slide 7: Definition: Bernoulli's Principle

ACS: AI.II.D.K1

Read the definition (PHAK 4-6). In the venturi tube, the air must speed up in the narrow throat to move the same amount of air in the same time, and the pressure there drops (PHAK 4-6).

Analogy: put your thumb over the end of a garden hose. The water speeds up where the opening narrows.

Source: PHAK 4-6, PHAK Fig 4-4

Slide 8: Newton's Third Law and Downwash

ACS: AI.II.D.K1

Newton's Third Law: "For every action, there is an equal and opposite reaction" (PHAK 4-3).

The downward backward flow from the top surface creates a downwash. Applying Newton's third law, the reaction to this downward backward flow is an upward forward force on the airfoil (PHAK 4-8).

Both Bernoulli's Principle and Newton's Laws are in operation whenever lift is being generated (PHAK 4-8). Figure 4-7 shows the pressure pattern changing as AOA changes.

Source: PHAK 4-3, PHAK 4-8, PHAK Fig 4-7

Slide 9: Definitions: Relative Wind and Angle of Attack

ACS: AI.II.D.K1

Read both definitions (PHAK G-25, PHAK G-3). The actual flight path determines the direction of the relative wind (PHAK G-25).

Figure 5-3: at high speed the AOA is small; at low speed the AOA must be larger to make the same lift.

Lift is perpendicular to the relative wind (PHAK 5-8), which your whiteboard notes also show.

Source: PHAK G-25, PHAK G-3, PHAK Fig 5-3

Slide 10: Lift, Speed, and AOA

ACS: AI.II.D.K1

Lift is proportional to the square of the velocity. An airplane at 200 knots has four times the lift of the same airplane at 100 knots, if AOA and other factors stay the same (PHAK 5-4).

The lift equation is L = CL x density x V squared x S, divided by 2 (PHAK 5-4). The pilot controls lift mainly by changing AOA and airspeed.

Figure 5-5 shows the coefficient of lift rising with AOA until it peaks, then falling off: that is the stall.

Source: PHAK 5-4, PHAK Fig 5-5

Slide 11: Definition: Critical Angle of Attack

ACS: AI.II.D.K1

Read the definition (PHAK G-9). When the critical AOA is exceeded, the airflow separates and lift drops: the airfoil has stalled (PHAK 5-7).

C172S example: power off at 2,550 lb, most rearward CG, flaps up: 48 KIAS wings level, 68 KIAS at 60 degrees of bank. Flaps full: 40 KIAS wings level (POH Fig 5-3).

Source: PHAK G-9, POH Fig 5-3

Slide 12: Quick Check

ACS: AI.II.D.K1

Quick check. Ask each question, wait for the answer, then confirm.

Question	Answer
State Bernoulli's Principle.	As the velocity of a moving fluid increases, the pressure within the fluid decreases (PHAK 4-6).
If you double your airspeed at the same AOA, what happens to lift?	It becomes four times as much (PHAK 5-4).
Can a wing stall at high airspeed?	Yes. It stalls at the critical AOA regardless of airspeed, attitude, or weight (PHAK G-9).

Transition: Next up, the four forces. The wing makes lift, but lift never comes alone. What else is acting on the airplane?

Source: PHAK 4-6, PHAK 4-8, PHAK 5-4, PHAK G-9

Slide 13: Forces Acting on an Airplane

ACS: Step 2 of 6 | AI.II.D.K4

Section start: Forces Acting on an Airplane. Goals: Thrust, drag, lift, weight; Forces in balance and in a climb; Parasite drag: form, interference, skin friction; Induced drag and L/D max.

Source: PHAK 5-1 to 5-8

Slide 14: The Four Forces

ACS: AI.II.D.K4

Thrust: the forward force produced by the powerplant/propeller. It opposes or overcomes drag (PHAK 5-1).

Drag: a rearward, retarding force caused by disruption of airflow by the wing, fuselage, and other protruding objects (PHAK 5-1).

Lift: produced by the dynamic effect of the air acting on the airfoil, acting perpendicular to the flight path through the center of lift (PHAK 5-1).

Weight: the combined load of the aircraft, crew, fuel, and cargo, acting vertically downward through the CG (PHAK 5-2).

Source: PHAK 5-1, PHAK 5-2, PHAK Fig 5-1

Slide 15: Forces in Balance and in a Climb

ACS: AI.II.D.K4

In steady flight, the sum of opposing forces is always zero. That does not mean the four forces are equal; it means the opposing forces cancel (PHAK 5-2).

In straight, level, unaccelerated flight, lift and weight are equal and are greater than thrust and drag, which are equal only to each other (PHAK 5-2).

Figure 5-2: in a climb, part of the weight acts rearward like drag, so more thrust is needed.

Source: PHAK 5-2, PHAK Fig 5-2

Slide 16: Definition: Parasite Drag

ACS: AI.II.D.K4

Read the definition (PHAK 5-6). The three types:

Form drag: the portion generated by the aircraft's shape and airflow around it, such as cowlings and antennas (PHAK 5-6). Your note "Jeep vs. Ferrari" is a good analogy.

Interference drag: comes from the intersection of airstreams that creates eddy currents, turbulence, or restricts smooth airflow, such as the wing root (PHAK 5-6). On the C172S, think of where the strut meets the wing.

Skin friction drag: aerodynamic resistance from moving air contacting the surface. Flush rivets, a smooth finish, and a clean, waxed airplane reduce it (PHAK 5-6, PHAK 5-7).

Source: PHAK 5-6, PHAK Fig 5-7, PHAK Fig 5-8

Slide 17: Induced Drag and the Drag Curve

ACS: AI.II.D.K4

Induced drag is inherent whenever an airfoil is producing lift and is inseparable from the production of lift (PHAK 5-7).

As AOA increases, induced drag increases. The lower the airspeed, the greater the AOA needed, so the greater the induced drag. Induced drag varies inversely with the square of the airspeed; parasite drag increases as the square of the airspeed (PHAK 5-8).

At L/D max, total drag is at its minimum (PHAK 5-5).

Note on your whiteboard: the PHAK says induced drag varies inversely with the square of airspeed and parasite drag increases as the square of airspeed, not just "inversely" and "directly."

Source: PHAK 5-7, PHAK 5-8, PHAK 5-5, PHAK Fig 5-6

Slide 18: Quick Check

ACS: AI.II.D.K4

Quick check. Ask each question, wait for the answer, then confirm.

Question	Answer
In straight and level flight, are all four forces equal?	No. Lift equals weight and thrust equals drag, but lift and weight are larger (PHAK 5-2).
Name the three types of parasite drag.	Form, interference, and skin friction (PHAK 5-6).
You slow down toward a stall. Which drag rises sharply?	Induced drag (PHAK 5-8).

Transition: Next up, wingtip vortices. Induced drag comes from air spilling around the wingtip. Where does that air go?

Source: PHAK 5-2, PHAK 5-6, PHAK 5-8

Slide 19: Wingtip Vortices

ACS: Step 3 of 6 | AI.II.D.K6

Section start: Wingtip Vortices. Goals: How vortices form; How they move and sink; Wake turbulence precautions.

Source: PHAK 5-7 to 5-10, AIM 7-4

Slide 20: How Wingtip Vortices Form

ACS: AI.II.D.K6

Below the wing the pressure is higher than above it. The air flows from the high pressure area below the tip up to the low pressure area on top, creating vortices that trail behind the airfoil (PHAK 5-7).

As the air curls upward around the tip, it combines with the downwash to form a fast-spinning trailing vortex (PHAK 5-9). Your whiteboard note matches this wording.

Vortex strength is directly proportional to weight and inversely proportional to wingspan and speed. Strongest during takeoff, climb, and landing (PHAK 5-9).

Source: PHAK 5-7, PHAK 5-8, PHAK 5-9, PHAK Fig 5-12

Slide 21: How Vortices Move

ACS: AI.II.D.K6

An aircraft makes vortices from the moment it rotates on takeoff until touchdown, since they are a by-product of lift (AIM 7-4-4).

Vortices from larger aircraft sink several hundred feet per minute, slowing and weakening with time and distance (AIM 7-4-4).

Within 100 to 200 feet of the ground, they move laterally at 2 or 3 knots. A light crosswind of 1 to 5 knots can hold the upwind vortex in the touchdown zone (PHAK Fig 5-15).

Vortices drift with the wind. A 10 knot wind moves them about 1,000 feet in a minute (PHAK 5-9).

Source: AIM 7-4-4, PHAK 5-9, PHAK Fig 5-15

Slide 22: Wake Turbulence Precautions

ACS: AI.II.D.K6

Landing behind a larger aircraft on the same runway: stay at or above its final approach path, note its touchdown point, and land beyond it (AIM 7-4-6).

Departing behind a larger aircraft: note its rotation point, rotate prior to it, and climb above its climb path until turning clear (AIM 7-4-6).

After a larger aircraft does a low approach, missed approach, or touch-and-go, wait at least 2 minutes before your takeoff or landing (AIM 7-4-6).

En route: avoid flight below and behind a large aircraft's path. Avoid following within 1,000 feet below (AIM 7-4-6, PHAK 5-9).

Whether or not ATC gives a wake turbulence caution, the pilot is expected to adjust to avoid wake encounters (AIM 7-4-6).

Source: AIM 7-4-6, PHAK 5-9

Slide 23: Quick Check

ACS: AI.II.D.K6

Quick check. Ask each question, wait for the answer, then confirm.

Question	Answer
When are wingtip vortices strongest?	When the generating aircraft is heavy, clean, and slow (PHAK 5-9).
Landing behind a larger aircraft on the same runway: where do you touch down?	Beyond its touchdown point, staying at or above its approach path (AIM 7-4-6).
A larger aircraft just did a touch-and-go. How long do you wait?	At least 2 minutes (AIM 7-4-6).

Transition: Next up, stability. We know the forces. Why does the airplane stay balanced when a gust hits it?

Source: PHAK 5-9, AIM 7-4-4, AIM 7-4-6

Slide 24: Stability, Maneuverability, Controllability

ACS: Step 4 of 6 | AI.II.D.K2

Section start: Stability, Maneuverability, Controllability. Goals: Static and dynamic stability; Pitch, roll, and yaw stability; Maneuverability and controllability; How CG changes them.

Source: PHAK 5-14 to 5-19

Slide 25: Definition: Stability

ACS: AI.II.D.K2

Read the definition (PHAK 5-14).

Static stability is the initial tendency to return to equilibrium. Positive: returns. Neutral: stays in the new condition. Negative: continues away (PHAK 5-14).

Source: PHAK 5-14, PHAK Fig 5-21

Slide 26: Dynamic Stability

ACS: AI.II.D.K2

Dynamic stability is the aircraft response over time. Positive: motion decreases. Neutral: neither decreases nor increases (like a worn shock absorber). Negative: motion increases (PHAK 5-15).

In flight demonstration: trim for hands-off level flight and give the controls a slight push. If the oscillations gradually stop, the airplane has positive stability; if they continue, neutral; if they increase, unstable (PHAK 5-17).

Source: PHAK 5-14, PHAK 5-15, PHAK 5-17, PHAK Fig 5-22

Slide 27: Longitudinal Stability (Pitch)

ACS: AI.II.D.K2

Longitudinal stability is stability about the lateral axis: pitch (PHAK 5-15).

It depends on the location of the wing and the horizontal tail relative to the CG, and on the size of the tail (PHAK 5-15).

Most airplanes have the center of lift behind the CG, so they are nose heavy. The horizontal stabilizer makes a downward force to balance it (PHAK 5-16).

If speed decreases, downwash on the tail decreases, the nose pitches down, and airspeed increases again. This is why the C172S tends to return to its trimmed speed (PHAK 5-16).

Source: PHAK 5-15, PHAK 5-16, PHAK Fig 5-24

Slide 28: Lateral Stability (Roll)

ACS: AI.II.D.K2

Lateral stability is stability about the longitudinal axis: roll. Four design factors: dihedral, sweepback, keel effect, and weight distribution (PHAK 5-17).

Dihedral: when a gust rolls the airplane, it sideslips. The wing slipping into the wind has a higher AOA and more lift, which rolls the airplane back (PHAK 5-18).

A high wing airplane like the C172S gets lateral stability from keel effect: the fuselage hangs below the wing like a pendulum (PHAK 5-18).

Source: PHAK 5-17, PHAK 5-18, PHAK G-10, PHAK Fig 5-29

Slide 29: Directional Stability (Yaw)

ACS: AI.II.D.K2

Directional stability is stability about the vertical axis: yaw. It is the most easily achieved stability in design (PHAK 5-19).

The vertical fin and fuselage sides aft of the CG make the airplane act like a weather vane or arrow, pointing into the relative wind. The farther aft and larger the fin, the greater the directional stability (PHAK 5-19).

Source: PHAK 5-19, PHAK Fig 5-32

Slide 30: Definitions: Maneuverability and Controllability

ACS: AI.II.D.K2

Read the definitions (PHAK 5-15). Maneuverability is governed by weight, inertia, size and location of the controls, structural strength, and powerplant. Controllability is the response to the pilot's control regardless of stability (PHAK 5-15).

Your whiteboard wording for maneuverability is close, but use the PHAK wording above on the checkride.

CG effects: forward CG stalls at a higher speed with higher elevator forces; aft CG is less stable and stall recovery gets harder (PHAK 5-43, PHAK 5-44).

Source: PHAK 5-15, PHAK 5-43, PHAK 5-44

Slide 31: Quick Check

ACS: AI.II.D.K2

Quick check. Ask each question, wait for the answer, then confirm.

Question	Answer
The nose returns toward level after a push, then the swings die out. Static and dynamic stability?	Positive static and positive dynamic (PHAK 5-14, PHAK 5-17).
Which axis is pitch stability about?	The lateral axis (PHAK 5-15).
What makes a C172S laterally stable?	Dihedral and keel effect from the high wing (PHAK

	5-18).

Transition: Next up, turning tendencies. The airplane is built stable, so what keeps pulling the nose to the left?

Source: PHAK 5-14 to 5-19

Slide 32: Turning Tendencies

ACS: Step 5 of 6 | AI.II.D.K3

Section start: Turning Tendencies. Goals: Torque reaction; Corkscrewing slipstream; Gyroscopic precession; P-factor.

Source: PHAK 5-30 to 5-32

Slide 33: Turning Tendencies: Torque Reaction

ACS: AI.II.D.K3

The left turning tendency called "torque" is made of four elements: torque reaction, corkscrewing slipstream, gyroscopic action, and P-factor (PHAK 5-30, PHAK 5-31).

Torque reaction uses Newton's Third Law: as the propeller turns one way, the airplane tries to roll the other way. Most U.S. engines turn the prop clockwise as seen from the pilot's seat (PHAK 5-31).

On the takeoff roll, more weight on the left main gear makes more friction and a left turning moment, corrected with rudder (PHAK 5-31).

Source: PHAK 5-30, PHAK 5-31, PHAK Fig 5-47

Slide 34: Turning Tendencies: Slipstream and Gyroscopic

ACS: AI.II.D.K3

Corkscrew effect: the prop gives the slipstream a spiraling rotation. At high power and low speed it strikes the vertical fin and yaws the airplane (PHAK 5-31).

Gyroscopic precession: a force applied to a spinning propeller takes effect 90 degrees ahead in the direction of rotation (PHAK 5-31). Most prominent in tailwheel airplanes when the tail is raised on takeoff, causing a left yaw (PHAK 5-32).

Source: PHAK 5-31, PHAK 5-32, PHAK Fig 5-48, PHAK Fig 5-49

Slide 35: Turning Tendencies: P-Factor

ACS: AI.II.D.K3

At a high AOA, the downward moving blade on the right has a greater "bite" than the upward moving blade. The center of thrust moves to the right of the prop disc, yawing the nose left (PHAK 5-32).

The four elements change with flight condition. The pilot must use the controls to compensate in every phase (PHAK 5-32).

Example: C172S: full power climb at Vy. Feel the nose wanting to go left and hold right rudder to keep the ball centered.

Source: PHAK 5-32, PHAK Fig 5-51

Slide 36: Quick Check

ACS: AI.II.D.K3

Quick check. Ask each question, wait for the answer, then confirm.

Question	Answer
Name the four turning tendencies.	Torque reaction, corkscrewing slipstream, gyroscopic precession, P-factor (PHAK 5-31).
Gyroscopic precession: where does the resulting force act?	90 degrees ahead in the direction of rotation (PHAK 5-31).
In a full power climb, which way does the nose yaw and what fixes it?	Left. Right rudder (PHAK 5-32).

Transition: Next up, load factors. Now that we can control it, how hard can we maneuver before it stalls or breaks?

Source: PHAK 5-31, PHAK 5-32

Slide 37: Load Factors

ACS: Step 6 of 6 | AI.II.D.K5

Section start: Load Factors. Goals: Load factor and design limits; Load factor in turns; Load factor and stall speed; Maneuvering speed.

Source: PHAK 5-33 to 5-36, POH 2-4, POH 2-11

Slide 38: Definition: Load Factor

ACS: AI.II.D.K5

Read the definition (PHAK G-18). A load factor of 3 means the total load on the structure is three times its weight (PHAK 5-33).

Load factors in aircraft design: designers find the highest load factors expected in normal operation, called limit load factors. The structure must support 1.5 times these without failure: the factor of safety (PHAK 5-33).

Category limits: normal 3.8 to -1.52, utility 4.4 to -1.76, acrobatic 6.0 to -3.00 (PHAK 5-34).

C172S: normal category flaps up +3.8g, -1.52g; flaps down +3.0g. Utility category flaps up +4.4g, -1.76g (POH 2-11).

Source: PHAK G-18, PHAK 5-33, PHAK 5-34, POH 2-11

Slide 39: Load Factor in Turns

ACS: AI.II.D.K5

In a coordinated, level turn, the load factor results from centrifugal force and weight (PHAK 5-34).

At 60 degrees of bank, load factor is 2 Gs. Load factor increases at a terrific rate after about 45 or 50 degrees (PHAK 5-34).

The yield factor of the average light plane is reached at about 70 to 75 degrees of bank (PHAK 5-36).

Source: PHAK 5-34, PHAK 5-36, PHAK Fig 5-53

Slide 40: Load Factor and Stall Speed

ACS: AI.II.D.K5

Stalling speed increases in proportion to the square root of the load factor. A 50 knot stalling speed becomes 100 knots at 4 Gs (PHAK 5-35).

C172S, flaps up, 2,550 lb: 48 KIAS at 0 degrees, 52 at 30, 57 at 45, and 68 at 60 degrees of bank (POH Fig 5-3).

Altitude loss during stall recovery may be as much as 230 feet (POH Fig 5-3).

Source: PHAK 5-35, POH Fig 5-3

Slide 41: Maneuvering Speed (VA)

ACS: AI.II.D.K5

VA is the speed below which you can move a single flight control, one time, to its full deflection, for one axis only, in smooth air, without risk of damage (PHAK 5-35).

Operating at or below VA does not provide structural protection against multiple full control inputs in one axis or full control inputs in more than one axis at the same time (PHAK 5-35).

C172S: 105 KIAS at 2,550 lb, 98 KIAS at 2,200 lb, 90 KIAS at 1,900 lb (POH 2-4).

Avoid rapid and large alternating control inputs; they may cause structural failure at any speed, including below VA (PHAK 5-36).

Source: PHAK 5-35, PHAK 5-36, POH 2-4

Slide 42: Quick Check

ACS: AI.II.D.K5

Quick check. Ask each question, wait for the answer, then confirm.

Question	Answer
Load factor in a level 60 degree bank?	2 Gs (PHAK 5-34).
C172S flaps up stall speed: wings level vs. 60 degrees of bank?	48 KIAS vs. 68 KIAS (POH Fig 5-3).
C172S VA at 2,550 lb, and does it protect against two full inputs at once?	105 KIAS. No (POH 2-4, PHAK 5-35).

Transition: Next up, risk management. Each principle comes with a risk. Let us manage them and plan how to teach this.

Source: PHAK 5-34, PHAK 5-35, POH 2-4, POH Fig 5-3

Slide 43: Risk Management and Teaching It

ACS: AI.II.D.R1, S1

Section start: Risk Management and Teaching It. Goals: Stalls from load factor; Wake, turning tendencies, spirals; Teaching three elements in flight.

Source: PHAK 5-20, PHAK 5-35, AIM 7-4-6

Slide 44: Risk: Stalls From Load Factor

ACS: AI.II.D.R1

Knowing load factors and VA helps prevent two of the most serious accident types: stalls from steep turns or excessive maneuvering near the ground, and structural failures from violent maneuvers or loss of control (PHAK 5-36).

Mitigation: plan the pattern to avoid overshoots, keep bank angles moderate near the ground, go around instead of tightening the turn, and reduce AOA at the first sign of a stall.

Source: PHAK 5-35, PHAK 5-36, POH Fig 5-3

Slide 45: Risk: Wake, Turning Tendencies, Spirals

ACS: AI.II.D.R1

Wake turbulence: pick a rotation point before a larger aircraft's, and wait 2 minutes after a larger aircraft low approach or touch-and-go (AIM 7-4-6).

Turning tendencies: anticipate left yaw on takeoff and in climbs; use right rudder to stay on centerline and coordinated (PHAK 5-32).

Spiral instability: in a spiral, pulling back before leveling the wings only tightens the turn and raises load factor. Improper recovery from spirals leading to structural failure has contributed to many general aviation fatalities (PHAK 5-20).

Source: AIM 7-4-6, PHAK 5-20, PHAK 5-32

Slide 46: Teaching It: Three Elements in Flight

ACS: AI.II.D.S1

S1 requires delivering instruction on at least three elements from K1 to K6. Example plan in the C172S at a safe altitude:

Stability (K2): trim for hands-off level flight, push the nose down slightly, release, and watch the oscillations damp out (PHAK 5-17).

Turning tendency (K3): in a full power climb, show the left yaw and the right rudder needed (PHAK 5-32).

Load factor (K5): fly 45 and 60 degree turns, feel the G load, and relate it to the higher stall speeds in the POH table (POH Fig 5-3).

Source: ACS AI.II.D.S1, PHAK 5-17, PHAK 5-32, POH Fig 5-3

Slide 47: Common Errors

ACS: AI.II.D

"All four forces are equal": opposing forces cancel, but lift and weight are larger than thrust and drag (PHAK 5-2).

"Stalls happen at low speed": stalls happen when the critical AOA is exceeded, at any airspeed (PHAK G-9).

"Induced drag varies inversely with speed": it varies inversely with the square of the airspeed (PHAK 5-8).

"Below VA I am safe": VA protects only against one full input in one axis (PHAK 5-35).

Source: PHAK 5-2, PHAK 5-8, PHAK G-9, PHAK 5-35

Slide 48: Review Questions

ACS: AI.II.D

Answers: (1) As velocity of a moving fluid increases, pressure decreases (PHAK 4-6). (2) Exceeding the critical AOA (PHAK G-9). (3) 2 Gs (PHAK 5-34). (4) Before the larger aircraft's rotation point, then climb above its path (AIM 7-4-6).

Source: PHAK 4-6, PHAK G-9, PHAK 5-34, AIM 7-4-6

Slide 49: References

ACS: AI.II.D

Review the references.

Source: See lesson plan for full list

7. Instructor Actions

- Follow the roadmap order and use each "Next up" box to transition.
- Draw the airfoil, four forces, and drag curve on the whiteboard as you go.
- Use the model airplane for the axes and stability demos.
- Ask the quick check questions after each section.
- Plan the S1 flight demonstrations for the next flight lesson.

8. Student Actions

- Follow along and draw the key diagrams.
- Answer the quick check questions.
- Explain one principle back in their own words.
- Identify where each principle shows up in the C172S.

9. Common Errors

Saying all four forces are equal. Opposing forces cancel, but lift and weight are larger than thrust and drag (PHAK 5-2).

Thinking stalls depend on airspeed. A wing stalls when it exceeds the critical AOA, regardless of airspeed, attitude, or weight (PHAK G-9).

Using "inversely" and "directly" for drag. Induced drag varies inversely with the square of airspeed; parasite drag increases as the square of airspeed (PHAK 5-8).

Believing VA protects against everything. Only one full input in one axis (PHAK 5-35).

Landing short of a larger airplane's touchdown point. Stay above its path and land beyond its touchdown point (AIM 7-4-6).

Not using right rudder in climbs. P-factor and slipstream yaw the nose left (PHAK 5-31, PHAK 5-32).

10. Conclusion and Review

Retrace the roadmap: the airfoil makes lift through Bernoulli and Newton; lift comes with the four forces and drag; induced drag leaves wingtip vortices; stability keeps the airplane in balance; turning tendencies pull it left; load factor limits how hard we can maneuver. Each risk comes straight from one of these principles.

Use the review questions, then have the student explain the whole chain back using the whiteboard.

11. Completion Standards

- Explains airfoil design characteristics and how lift is made (AI.II.D.K1).
- Explains stability, maneuverability, and controllability (AI.II.D.K2).
- Explains the four turning tendencies (AI.II.D.K3).
- Explains the four forces and types of drag (AI.II.D.K4).
- Explains load factors in airplane design and their effect on stall speed (AI.II.D.K5).
- Explains wingtip vortices and the avoidance procedures (AI.II.D.K6).
- Identifies and mitigates the risks of these principles (AI.II.D.R1).
- Delivers instruction on at least three elements from K1 to K6 (AI.II.D.S1).

12. References

- FAA-S-ACS-25, Flight Instructor for Airplane Category ACS (November 2023), Area II, Task D.
- FAA-H-8083-25C, Pilot's Handbook of Aeronautical Knowledge (PHAK): Chapter 4 (4-6 to 4-10), Chapter 5 (5-1 to 5-20, 5-30 to 5-36, 5-43, 5-44), and Glossary (G-1, G-3, G-6, G-9, G-10, G-18, G-25). The October 2025 addendum does not change these pages.
- Aeronautical Information Manual (AIM) with Changes 1 to 3: 7-4-4 Vortex Behavior and 7-4-6 Vortex Avoidance Procedures.
- Cessna 172S NAV III (G1000) POH pages, P/N 172SPHAUS, from Information Manual 172SIMAUS Rev 3: 2-4 (Figure 2-1), 2-11, 5-12 (Figure 5-3). Training use only.
- FAA-H-8083-9, Aviation Instructor's Handbook: 5-9 (lesson introduction).
- Applicant's own whiteboard notes (Principles of Flight Characteristics) were used as the lesson outline and checked against the PHAK.